

Convergence of different ADC(n) variants through $n = 15$

These notes builds on the Dyson vs non-Dyson controversy, and assesses how and if the different ways, known to the literature, to generate a systematic ADC converge to the FCI limit.

I. THE λ -SCALED PROBLEM

Sector I are the $(N - 1)$ configurations (2h1p, ...), sector II the $(N + 1)$ ones (2p1h, ...). All results are properties of the hamiltonian

$$\hat{H}(\lambda) = \hat{F} + \lambda(\hat{H} - \hat{F}), \quad (1)$$

where

$$\hat{F} = \sum_p \epsilon_p \hat{a}_p^\dagger \hat{a}_p \quad (2)$$

is the canonical HF Fock operator, built as dense matrices in the determinant spaces $\mathcal{H}_N$ and $\mathcal{H}_{N\pm 1}$ of the systems in Table I. FCI gives the exact $\Sigma(\omega; \lambda)$ at any λ ; its Taylor coefficients follow from complex λ on a circle of radius $r = 0.3$ with $N_\lambda = 32$ points,¹

$$\Sigma^{(m)}(\omega) = \frac{1}{N_\lambda r^m} \sum_{j=0}^{N_\lambda-1} \Sigma(\omega; r e^{2\pi i j / N_\lambda}) e^{-2\pi i m j / N_\lambda}. \quad (3)$$

The only perturbative input to ADC is the Rayleigh-Schrödinger (RS) series of the ground state and its energy,

$$|\Psi_0(\lambda)\rangle = \sum_{m \geq 0} \lambda^m |\Psi_0^{(m)}\rangle, \quad (4)$$

$$E_0(\lambda) = \sum_{m \geq 0} \lambda^m E_0^{(m)}. \quad (5)$$

The zeroth-order state is the HF determinant $|\Phi_0\rangle$, which is an eigenstate of $\hat{F}$,

$$\hat{F} |\Phi_0\rangle = E_0^{(0)} |\Phi_0\rangle, \quad (6)$$

with $E_0^{(0)}$ the sum of the occupied orbital energies. We use intermediate normalization, *i.e.* every correction is orthogonal to the HF determinant,

$$\langle \Phi_0 | \Psi_0^{(m)} \rangle = \delta_{m0}. \quad (7)$$

Inserting the two series into the Schrödinger equation of $\hat{H}(\lambda)$ and collecting the terms of order λ^m gives the recursion

$$|\Psi_0^{(m)}\rangle = \hat{R}_0 \left[(\hat{H} - \hat{F}) |\Psi_0^{(m-1)}\rangle - \sum_{l=1}^m E_0^{(l)} |\Psi_0^{(m-l)}\rangle \right]. \quad (8)$$

Here $\hat{R}_0$ is the reduced resolvent of $\hat{F}$,

$$\hat{R}_0 = \hat{Q} (E_0^{(0)} - \hat{F})^{-1} \hat{Q}, \quad (9)$$

where

$$\hat{Q} = 1 - |\Phi_0\rangle \langle \Phi_0| \quad (10)$$

projects out the HF determinant. Because $\hat{F}$ is diagonal in the determinant basis, applying $\hat{R}_0$ amounts to dividing each determinant coefficient by the difference between $E_0^{(0)}$ and the orbital-energy sum of that determinant. The projector keeps the corrections orthogonal to $|\Phi_0\rangle$ and removes the one vanishing denominator. The energy corrections follow from projecting the order- l equation onto $|\Phi_0\rangle$,

$$E_0^{(l)} = \langle \Phi_0 | \hat{H} - \hat{F} | \Psi_0^{(l-1)} \rangle. \quad (11)$$

Since the matrix elements of Sec. II need the normalized state, every series is finally divided by the series of $\langle \Psi_0 | \Psi_0 \rangle$.

TABLE I. Systems (STO-3G). n_{occ} , n_{vir} : occupied and virtual spin orbitals; k_{max} : last non-empty class; $n_c = 2k_{\text{max}}$: first order with the complete manifold; n_{op} : size of the complete manifold.

system	geometry (Å)	n_{occ}	n_{vir}	k_{max}	n_c	n_{op}
H ₄ chain	spacing 1.0	4	4	3	6	112
H ₄ chain	spacing 1.6	4	4	3	6	112
LiH	1.6	4	8	4	8	1012
H ₆ chain	spacing 1.0	6	6	5	10	1584

II. INTERMEDIATE-STATE BLOCKS

The excitation operators of the two sectors are the ionizing strings

$$\{\hat{r}_v^-\} = \{\hat{a}_b^\dagger \hat{a}_j \hat{a}_i, \hat{a}_c^\dagger \hat{a}_b^\dagger \hat{a}_k \hat{a}_j \hat{a}_i, \dots\} \quad (12)$$

of sector I and the attaching strings

$$\{\hat{r}_v^+\} = \{\hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j, \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_c^\dagger \hat{a}_k \hat{a}_j, \dots\} \quad (13)$$

of sector II. The ADC supermatrix is labelled by the single-character manifold

$$\{\hat{X}_J\} = \{\hat{a}_p^\dagger\} \cup \{\hat{r}_v^+\} \cup \{\hat{r}_v^-\}, \quad (14)$$

in which every operator adds one electron. It is ordered into classes k : $\hat{a}_p^\dagger$ is class 0; class k holds the $(k+1)$ pkh strings of sector II and the adjoints of the $(k+1)$ hpk strings of sector I. Each $\hat{X}_J$ becomes the vector pair $(\hat{X}_J \Psi_0, \hat{X}_J^\dagger \Psi_0)$ in $\mathcal{H}_{N+1} \oplus \mathcal{H}_{N-1}$, so that

$$\mathbb{S}_{IJ} = \langle \hat{X}_I \Psi_0 | \hat{X}_J \Psi_0 \rangle + \langle \hat{X}_J^\dagger \Psi_0 | \hat{X}_I^\dagger \Psi_0 \rangle \quad (15)$$

is the metric $\langle \{\hat{X}_I^\dagger, \hat{X}_J\} \rangle$ and

$$\mathbb{M}_{IJ} = \langle \hat{X}_I \Psi_0 | \hat{H} - E_0 | \hat{X}_J \Psi_0 \rangle - \langle \hat{X}_J^\dagger \Psi_0 | \hat{H} - E_0 | \hat{X}_I^\dagger \Psi_0 \rangle \quad (16)$$

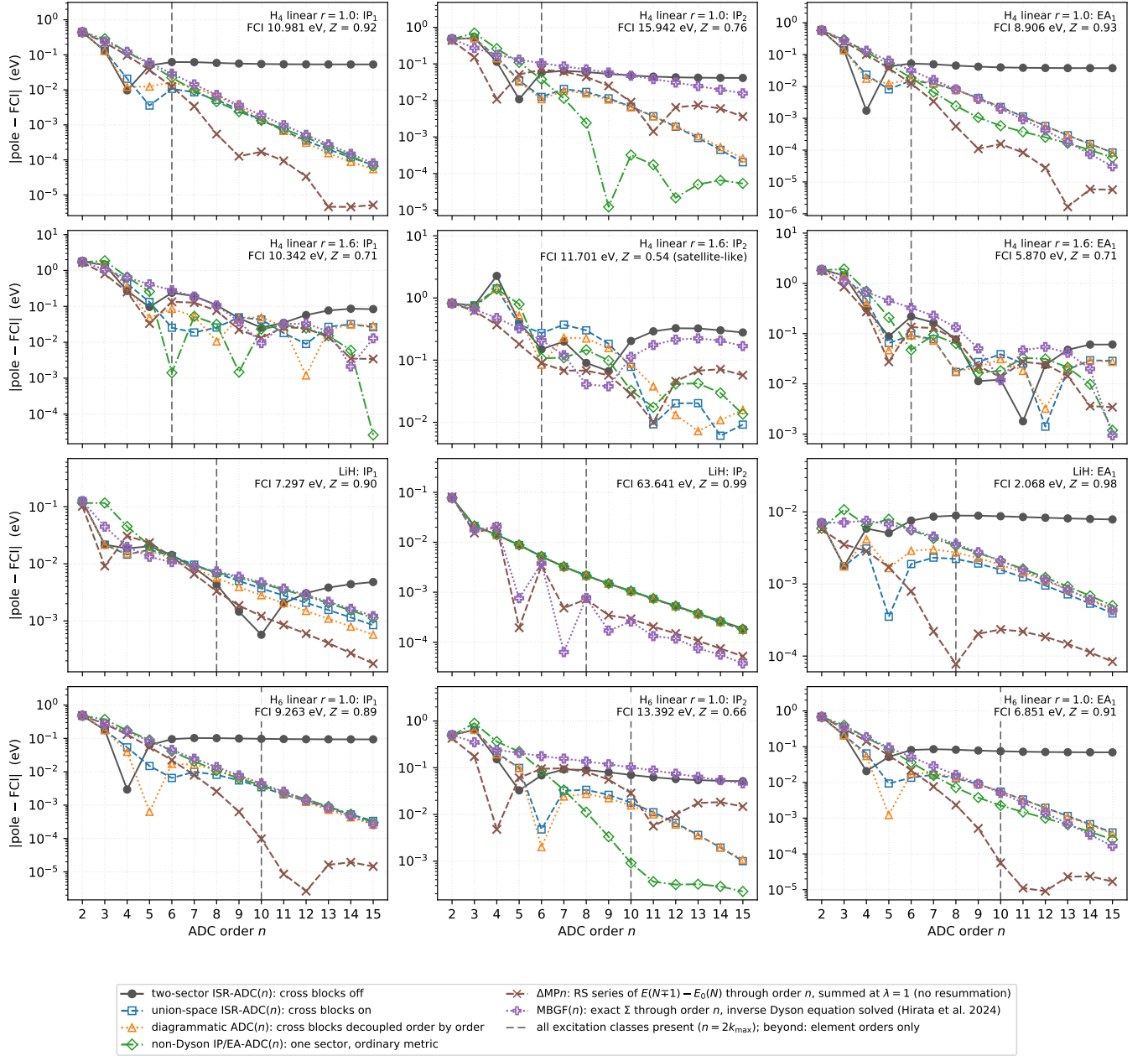


FIG. 1. Principal ADC(n) poles against FCI at $\lambda = 1$. Dashed line: $n = 2k_{\max}$, beyond which the manifold is complete (for LiH the non-Dyson IP sector is complete at $n = 6$).

is the superoperator $\langle \{\hat{X}_I^\dagger, [\hat{H}, \hat{X}_J]\} \rangle$. Inserting the RS series gives $\mathbb{S}^{(m)}$ and $\mathbb{M}^{(m)}$ exactly. The intermediate states²⁻⁴ follow from Gram-Schmidt of class k against all lower classes and Löwdin orthonormalization within it, both done on the power series. The orthonormal supermatrix is

$$\tilde{\mathbb{M}} = \begin{pmatrix} \epsilon + \Sigma(\infty) & \mathbf{U}^I & \mathbf{U}^{II} \\ \mathbf{U}^{I\dagger} & -\mathbf{K}^I - \mathbf{C}^I & \mathbf{X} \\ \mathbf{U}^{II\dagger} & \mathbf{X}^\dagger & \mathbf{K}^{II} + \mathbf{C}^{II} \end{pmatrix}, \quad (17)$$

where $\mathbf{K}$ is the diagonal zeroth order and $\mathbf{X}$ the cross block between the sectors (its metric counterpart vanishes identically).

The inner projection⁵ $\mathbf{G} = [\mathbb{S}(\omega\mathbb{S} - \tilde{\mathbb{M}})^{-1}\mathbb{S}]_{00}$ then has the Dyson form with

$$\Sigma(\omega) = \Sigma(\infty) + \mathbf{U}(\omega - \tilde{\mathbb{M}}_{\text{cc}})^{-1}\mathbf{U}^\dagger, \quad (18)$$

$\tilde{\mathbb{M}}_{\text{cc}}$ being all classes $k \geq 1$. Over the complete manifold this reproduces FCI to 10^{-13} . Folded order by order, the dynamic part $\Sigma_{\text{dyn}} = \Sigma - \Sigma(\infty)$ is

$$\Sigma_{\text{dyn}}^{(m)}(\omega) = \sum_{a+b+c=m} \mathbf{U}^{(a)} \mathbf{R}^{(b)}(\omega) \mathbf{U}^{(c)\dagger}, \quad (19)$$

where $\mathbf{R}^{(b)}$ is the order- b coefficient of $(\omega - \tilde{\mathbb{M}}_{\text{cc}})^{-1}$. This is the exact Taylor reference; it agrees with the contour coefficients

to 5×10^{-13} . $\text{ADC}(n)^{6,7}$ keeps classes $k \leq \lfloor n/2 \rfloor$ and block (k, l) through order $n - k - l$,

$$[\tilde{\mathbb{M}}_{\text{ADC}(n)}]_{kl} = \sum_{m=0}^{n-k-l} \tilde{\mathbb{M}}_{kl}^{(m)}. \quad (20)$$

Its eigenvalues at $\lambda = 1$ are the poles; principal poles have orbital weight $Z > 0.3$. For $n \geq n_c$ (Table I) the manifold is complete and higher n only adds orders to its matrix elements.

III. CONSTRUCTIONS

All Dyson variants share $\Sigma(\infty)$, $\mathbf{U}$ and the same-sector blocks exactly; they differ in $\mathbf{X}$.

TABLE II. The constructions of Fig. 1. Limit: complete manifold, all orders.

construction	$\mathbf{X}$	exact through	limit
two-sector $\text{ADC}(n)$	dropped	λ^3	not FCI
union $\text{ADC}(n)$	Eq. (20)	λ^n	FCI
diagrammatic $\text{ADC}(n)$	rotated	λ^n	FCI
non-Dyson $\text{ADC}(n)$	none	λ^n	FCI
$\Delta\text{MP}n$	–	λ^n	RS series
$\text{MBGF}(n)$	–	Σ to λ^n	Dyson root

A. Union-space ISR- $\text{ADC}(n)$

Union-space ISR- $\text{ADC}(n)$ keeps $\mathbf{X}$ under the rule of Eq. (20). The block between class k of sector I and class l of sector II starts at order

$$L_{kl} = \left\lceil \frac{k + l + 1}{2} \right\rceil, \quad (21)$$

so under the rule the block $\mathbf{X}_{11}$ enters at $n = 4$, $\mathbf{X}_{12}$ at $n = 5$, $\mathbf{X}_{22}$ and $\mathbf{X}_{13}$ at $n = 7$, $\mathbf{X}_{23}$ at $n = 8$ and $\mathbf{X}_{33}$ at $n = 10$. With these blocks the fold is exact through λ^n at every n tested ($n = 4$ –9, to 10^{-12}).

B. Two-sector ISR- $\text{ADC}(n)$ and the cross block

Two-sector ISR- $\text{ADC}(n)$ orthonormalizes and folds each sector on its own, and thus yields the familiar form of the self-energy, organized by + and – branches,

$$\Sigma(\omega) = \Sigma(\infty) + \mathbf{U}^{\text{II}}(\omega - \mathbf{K}^{\text{II}} - \mathbf{C}^{\text{II}})^{-1} \mathbf{U}^{\text{II}\dagger} + \mathbf{U}^{\text{I}}(\omega + \mathbf{K}^{\text{I}} + \mathbf{C}^{\text{I}})^{-1} \mathbf{U}^{\text{I}\dagger}. \quad (22)$$

Its blocks are exactly the same-sector blocks of $\tilde{\mathbb{M}}$, so it is $\tilde{\mathbb{M}}$ with $\mathbf{X}$ deleted. The cross block vanishes at the two lowest orders,

$$\mathbf{X}^{(0)} = \mathbf{X}^{(1)} = 0, \quad (23)$$

but not at second, where it is a contraction of the first-order doubles with the two-electron integrals. Expanding the union

fold in $\mathbf{X}$, the leading term the two-sector construction misses is

$$\Delta\Sigma^{(4)} = \mathbf{U}^{\text{I}(1)}(\omega + \mathbf{K}_1^{\text{I}})^{-1} \mathbf{X}_{11}^{(2)}(\omega - \mathbf{K}_1^{\text{II}})^{-1} \mathbf{U}^{\text{II}(1)\dagger} + \text{h.c.}, \quad (24)$$

which matches the λ^4 difference of the two folds to 5×10^{-18} . Two-sector ISR- $\text{ADC}(n)$ is therefore exact only through λ^3 , for every n .

The defect is fundamental, its complete-manifold limit is not FCI, as can be seen clearly in Fig. 1. This obstacle has been identified by Tarantelli and Cederbaum.⁵ They write the Dyson equation as a configuration-interaction problem in $\mathcal{H}_{N+1} \oplus \mathcal{H}_{N-1}$, with a composite Hamiltonian that is $\mathbb{M}$ of Eq. (16). The block that couples the $(N + 1)$ and $(N - 1)$ configurations vanishes in zeroth and first order but not in second, and they conclude that this coupling “prevents the possibility of reproducing analytically the fourth-order ADC scheme”, since ADC assumes that the self-energy splits into independent $(N + 1)$ and $(N - 1)$ parts. They remove the coupling by a further unitary transformation that block-diagonalizes the configuration space. This transformation can only be built order by order, and with an exactly solvable two-electron example they show that no transformation built from the ground state alone achieves the decoupling. The two-sector ISR is built from the ground state alone and is decoupled by construction.

C. Diagrammatic $\text{ADC}(n)$

Diagrammatic $\text{ADC}(n)$ removes $\mathbf{X}$ by the Tarantelli–Cederbaum rotation⁵ $\mathbf{W} = e^{\mathbf{A}}$, with $\mathbf{A}$ antisymmetric in the I–II blocks and

$$A_{\mu\nu}^{(m)} = \frac{R_{\mu\nu}^{(m)}}{K_\nu - K_\mu}, \quad (25)$$

$\mathbf{R}^{(m)}$ being the I–II part of $[\mathbf{W}^\dagger \tilde{\mathbb{M}} \mathbf{W}]^{(m)}$ from $\mathbf{A}^{(<m)}$; Eq. (20) is applied to the rotated blocks. At third order this moves $\mathbf{X}^{(2)}$ into the couplings,

$$\delta U_{pv}^{\text{II}(3)} = \sum_{\mu \in \text{I}_1} \frac{U_{p\mu}^{\text{I}(1)} X_{\mu\nu}^{(2)}}{K_\nu - K_\mu}, \quad (26)$$

where $K_\nu - K_\mu$ is a 3p3h energy; these rows reproduce the diagrammatic $\text{ADC}(4)$ rows⁷ to 10^{-18} .

D. Non-Dyson $\text{ADC}(n)$

Non-Dyson $\text{ADC}(n)^{8,9}$ builds the ISR from $\hat{r}_v^- \Psi_0$ in $\mathcal{H}_{N-1}$ (or $\hat{r}_v^+ \Psi_0$ in $\mathcal{H}_{N+1}$) with the ordinary overlap and the same rule; its eigenvalues are the IPs (EAs).

E. Perturbative references

$\Delta\text{MP}n$ is the RS series of $E_{N-1}(\lambda) - E_0(\lambda)$ for the Koopmans-connected state, summed through order n . $\text{MBGF}(n)^{10}$ takes

the exact $\Sigma^{[n]} = \sum_{m \leq n} \Sigma^{(m)}$ and solves

$$\omega = e_p(\omega), \quad (27)$$

with e_p the eigenvalue of $\epsilon + \Sigma^{[n]}(\omega)$ of largest weight on p , taking the root of largest residue. The pole Taylor series of union, diagrammatic and non-Dyson ADC(n) equal the exact one through λ^n to 10^{-17} ($n \leq 15$); the two-sector pole deviates from λ^4 on.

IV. NE AND AR BEYOND ADC(5)

Fig. 2 follows the non-Dyson ADC(n) series of the Ne and Ar valence ionizations in correlation-consistent bases to high order, against the FCI IP of the same block. The series are the exact symmetry-adapted ones of the λ -scaled problem of the previous sections, so no approximation enters beyond the basis: Ne/cc-pVDZ (1s frozen) through $n = 9$, Ne/aug-cc-pVDZ (1s frozen) and Ar/aug-cc-pVDZ (1s2s2p frozen) through $n = 7$. ADC(1) is Koopmans' theorem. The lower row zooms on $n \geq 5$. The error of the Ne/aug-cc-pVDZ series is still +0.09 and +0.13 eV at $n = 7$ for 2p and 2s: in a diffuse basis the series itself converges slowly, and this is what the ADC(5) errors of Figs. 3–5 contain.

V. DYSON ADC(2)–(5) FOR MOLECULES

Figs. 3–5 show the Dyson diagrammatic ADC(n) IP for $n = 2$ –5 for the states of the closed-shell molecules for which

an ADC(5) value exists (frozen core, the convention of the reference data; aug-cc-pVDZ). The error is taken against the selected CI (sCI) value of the same basis, so that no basis increment enters; the state assigned to each level is fixed from the ADC(4) calculation of the molecule. The grey line is CC2, CCSD, CC3 and CCSDT. The ADC(5) errors are not small: the mean absolute ADC(5) error of the 24 states shown at the time of writing is 0.34 eV, against 0.04 eV for CCSDT, and it is largest for BeO and BN. For most states the error alternates in sign with n and does not decrease monotonically.

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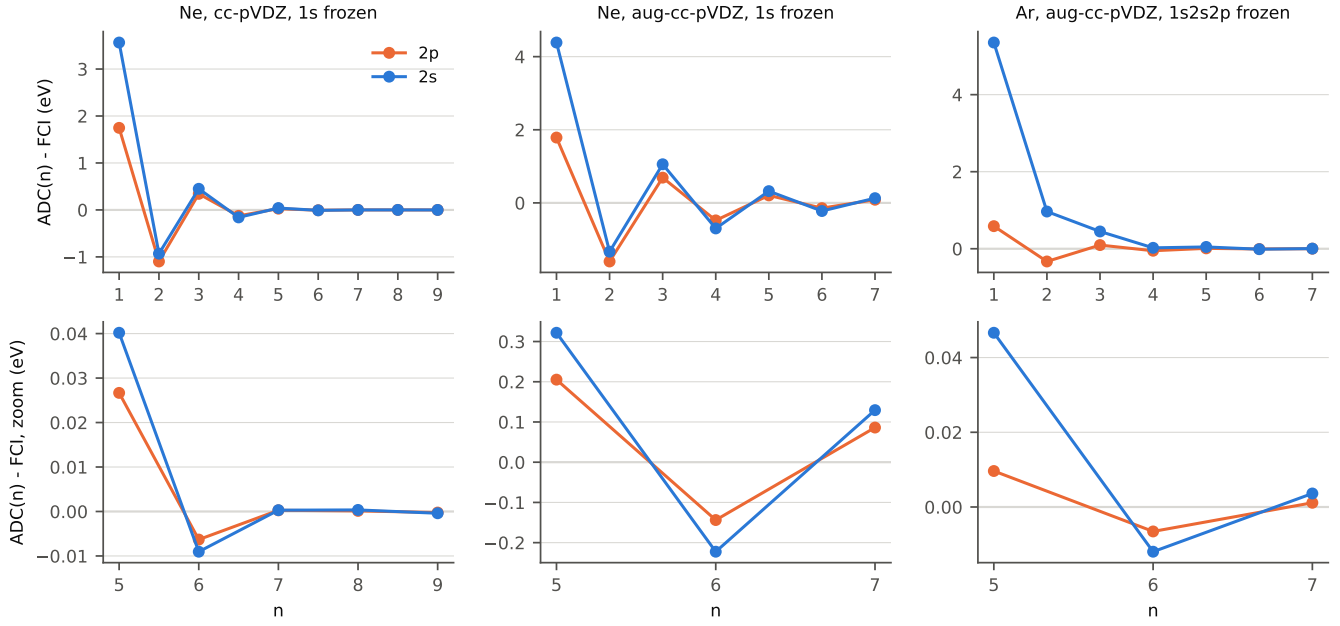
Non-Dyson ADC(n) against FCI, exact oracle

FIG. 2. Non-Dyson ADC(n) IP minus the FCI IP of Ne/cc-pVDZ, Ne/aug-cc-pVDZ and Ar/aug-cc-pVDZ (frozen core as stated in the text). Top: $n = 1$ (Koopmans) to the last order; bottom: $n \geq 5$.

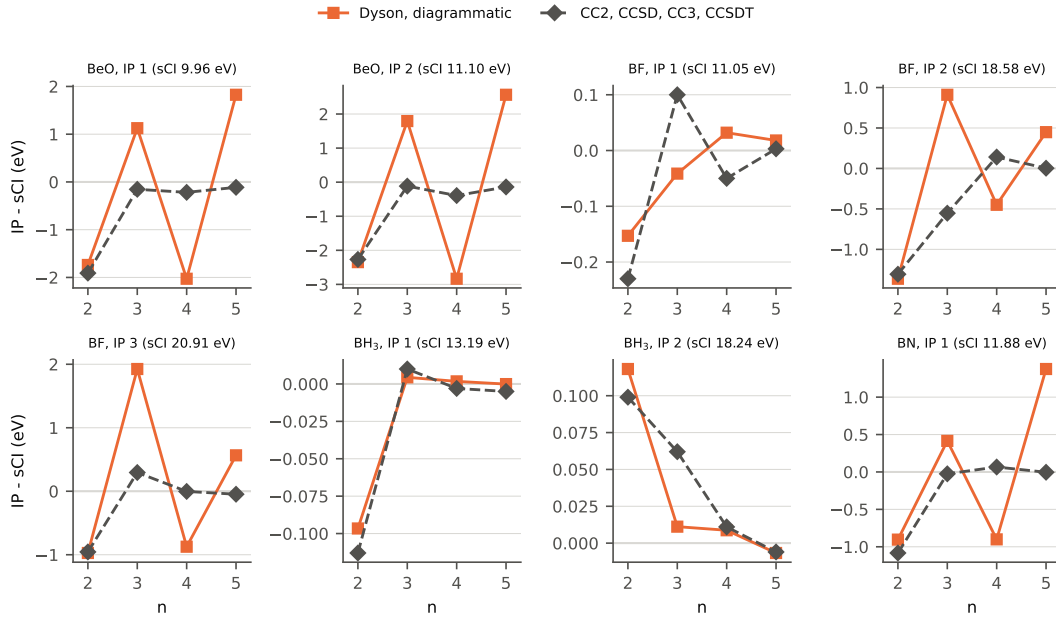


FIG. 3. Dyson diagrammatic ADC(n) IP and coupled cluster minus the sCI IP, frozen core, aug-cc-pVDZ (1/3).

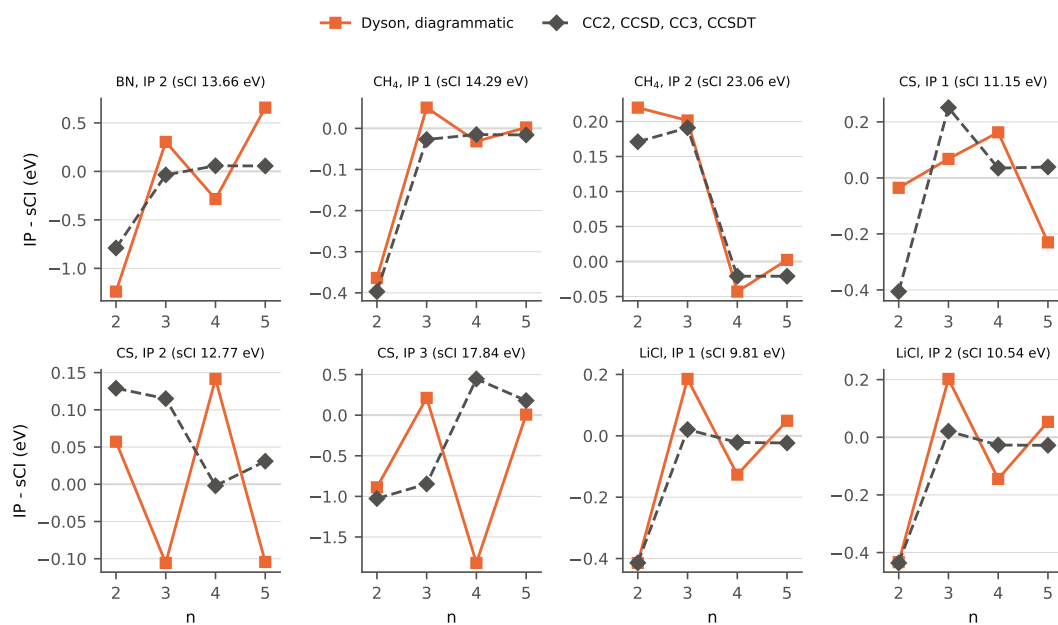


FIG. 4. As Fig. 3 (2/3).

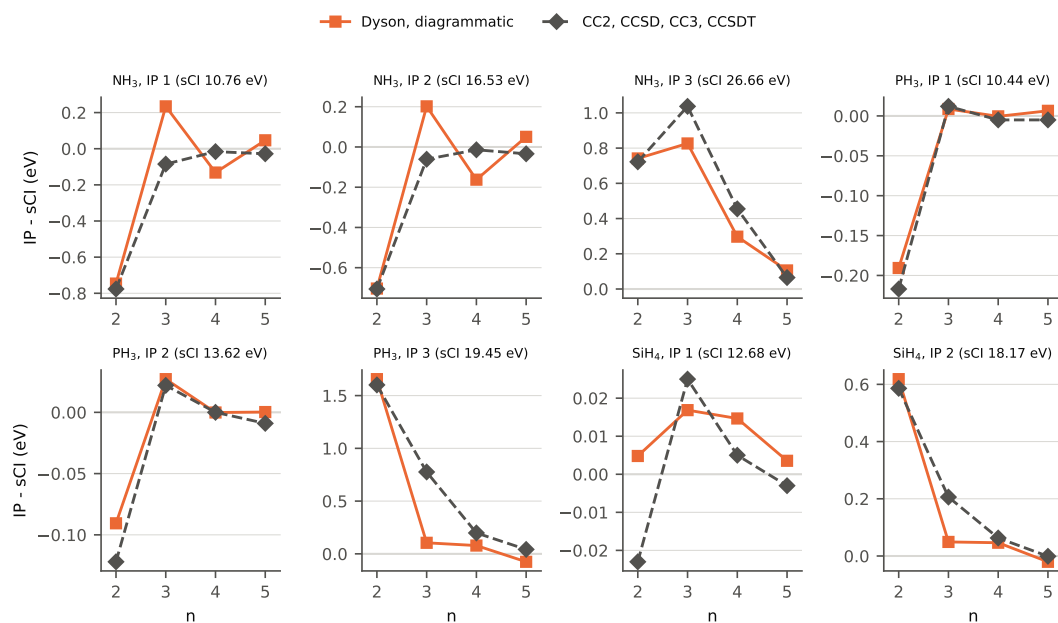


FIG. 5. As Fig. 3 (3/3).